

Date: Oct 01, 2025

# Math Trimester 1

MorningWork

Name:

Class of:

Teacher:

Homerroom:

## Multiple Choice — Quadratic Functions

1) Which of the following is a quadratic function?

- A.  $f(x) = 3x + 5$
- B.  $f(x) = x^2 - 4x + 1$
- C.  $f(x) = \sqrt{x} + 2$
- D.  $f(x) = \frac{4}{x}$

2) The graph of  $y = (x - 3)^2 - 7$  has its vertex at:

- A.  $(3, 7)$
- B.  $(-3, -7)$
- C.  $(3, -7)$
- D.  $(-3, 7)$

3) What are the zeros (x-intercepts) of  $y = x^2 - 5x + 6$ ?

- A.  $x = -2, -3$
- B.  $x = 2, 3$
- C.  $x = -2, 3$
- D.  $x = -1, 6$

4) For the quadratic  $y = -2(x + 1)^2 + 9$ , the parabola:

- A. opens upward and has a minimum at  $y = 9$
- B. opens upward and has a maximum at  $y = 9$
- C. opens downward and has a maximum at  $y = 9$
- D. opens downward and has a minimum at  $y = 9$

5) Consider  $y = ax^2 + bx + c$ . If the discriminant  $b^2 - 4ac$  is negative, which statement is true?

- A. The graph has two distinct real x-intercepts.
- B. The graph has one real x-intercept (a repeated root).
- C. The graph has no real x-intercepts.
- D. The graph is not a parabola.

## Quadratic Functions — MCQ Solutions

1) Which of the following is a quadratic function?

- A.  $f(x) = 3x + 5$
- B.  $f(x) = x^2 - 4x + 1$
- C.  $f(x) = \sqrt{x} + 2$
- D.  $f(x) = \frac{4}{x}$

**Solution:** A quadratic function has the form  $ax^2 + bx + c$  with  $a \neq 0$ .  $A$  is linear (degree 1),  $C$  involves a square root (not a polynomial), and  $D$  is a reciprocal (rational, not polynomial). Only  $B$  has degree 2.  $\Rightarrow$  **Answer: B.**

2) The graph of  $y = (x - 3)^2 - 7$  has its vertex at:

- A.  $(3, 7)$
- B.  $(-3, -7)$
- C.  $(3, -7)$
- D.  $(-3, 7)$

**Solution:** In vertex form  $y = (x - h)^2 + k$ , the vertex is  $(h, k)$ . Here  $h = 3$  and  $k = -7$ , so the vertex is  $(3, -7)$ .  $\Rightarrow$  **Answer: C.**

3) What are the zeros (x-intercepts) of  $y = x^2 - 5x + 6$ ?

- A.  $x = -2, -3$
- B.  $x = 2, 3$
- C.  $x = -2, 3$
- D.  $x = -1, 6$

**Solution:** Set  $y = 0$ :  $x^2 - 5x + 6 = 0$ . Factor:  $(x - 2)(x - 3) = 0$ . Thus  $x = 2$  or  $x = 3$ .  $\Rightarrow$  **Answer: B.**

4) For the quadratic  $y = -2(x + 1)^2 + 9$ , the parabola:

- A. opens upward and has a minimum at  $y = 9$
- B. opens upward and has a maximum at  $y = 9$
- C. opens downward and has a maximum at  $y = 9$
- D. opens downward and has a minimum at  $y = 9$

**Solution:** In  $y = a(x - h)^2 + k$ , the sign of  $a$  determines opening:  $a < 0 \Rightarrow$  opens downward; vertex is  $(h, k)$ . Here  $a = -2 < 0$ ,  $h = -1$ ,  $k = 9$ . So it opens downward with a maximum value 9 at  $x = -1$ .  $\Rightarrow$  **Answer: C.**

5) Consider  $y = ax^2 + bx + c$ . If the discriminant  $b^2 - 4ac$  is negative, which statement is true?

- A. The graph has two distinct real x-intercepts.
- B. The graph has one real x-intercept (a repeated root).
- C. The graph has no real x-intercepts.
- D. The graph is not a parabola.

**Solution:** The discriminant  $\Delta = b^2 - 4ac$  determines the nature of roots. If  $\Delta < 0$ , there are no real solutions—only complex conjugate roots—so the parabola does not cross the  $x$ -axis.  $\Rightarrow$  **Answer: C.**